

Optics became clearer — Solution

X26 (2026) — English translation

A1

0.60 pts

For 5 different values of a , obtain 5 corresponding values of h at which the LED image S' is focused on the screen. For each measurement, calculate the focal length value f . Average the resulting values f and write down the average value f_{avg} on your answer sheet.

a , mm b , mm f , mm 210 828 167.5 225 694 169.9 235 620 170.4 252 528 170.5 280 429 169.4

Answer: \textbf{Answer:}

$$f_{\text{avg}} = (170 \pm 2) \text{ mm}$$

B0

0.10 pts

Measure and write down the distance a_0 on your answer sheets.

Answer: \textbf{Answer:}

$$a_0 = (207 \pm 2) \text{ mm}$$

B1

0.40 pts

By changing the position of the d aperture, observe the changes in the image on the screen. Mark ($\checkmark$) all correct statements on your answer sheets.

As the aperture approaches the source, the brightness of the image increases; as the aperture moves away from the source, its brightness decreases.

Answer: \textbf{Answer:} The size of the image and its shape remain unchanged. As the aperture approaches the source, the brightness of the image increases; as the aperture moves away from the source, its brightness decreases.

B2

0.60 pts

For 8 different distances d between the source and aperture D_2 , measure the dimensions m'_1 , m'_2 of both diagonals of the cross image on the screen. Vary d as much as possible. Calculate for each d the average image size m' .

d , mm m'_1 , mm m'_2 , mm $1/d$, m^{-1} m' , mm 50 88 88 20.0 ± 0.80 88 75 88 88 13.3 ± 0.36 88 100 88 88 10.0 ± 0.20 88 125 88 88 8.00 ± 0.13 88 143 85 85 6.99 ± 0.10 85 160 64 64 6.25 ± 0.08 64 182 55 55 5.49 ± 0.06 55 204 49 49 4.90 ± 0.05 49 226 43 43 4.42 ± 0.04 43 244 40 40 4.10 ± 0.03 40 262 35 35 3.82 ± 0.03 35 287 32 32 3.48 ± 0.02 32 321 28 28 3.12 ± 0.02 28 350 27 27 2.86 ± 0.02 27 ± 2 ± 1 ± 1 ± 1

B3

1.50 pts

Obtain the theoretical dependence $m'(d)$ via m , d , f , a_0 . Plot a linearized graph and obtain the size m of the cross.

Let us highlight with a dotted line a plane at a distance b from the source, which is focused by the lens onto the screen. It is obvious that from this plane only rays limited by the dotted line will reach the screen. Then we can consider an image of size m' on the

screen as an image of a segment of length l in the indicated plane.

$$l = \frac{m}{d} \cdot b$$

Then смер изображения на экране:

$$m' = l \cdot \frac{h - a_0 - f}{a_0 + f - b}$$

Formula тонкой линзы:

$$\frac{1}{f} = \frac{1}{a_0 + f - b} + \frac{1}{h - a_0 - f}$$

Hence:

$$a_0 + f - b = f \cdot \frac{h - a_0 - f}{h - a_0 - 2f}$$

Excluding b we get:

$$m' = -\frac{m}{fd}(h - a_0 - 2f) \cdot (f \cdot \frac{h - a_0 - f}{h - a_0 - 2f} - a_0 - f)$$

□13□Substituting $h - a_0$ from the formula тонкой линзы:

$$h - a_0 = \frac{a_0 f}{a_0 - f}$$

$$m' = \frac{m}{d} \cdot \left(\frac{a_0}{a_0 - f}(2f - a_0) - f \right) = \frac{m}{d} \cdot \left(\frac{a_0 f}{a_0 - f} - (a_0 + f) \right)$$

Линеаризация:

$$m'(1/d)$$

Dots which lie on the plateau - aperture by the lens; we will not plot them on the graph, because they are not needed to calculate the slope factor.

From the resulting graph we learn the slope coefficient of the line:

$$k_1 = (1.21 \pm 0.06) \cdot 10^{-2} \text{ m}^2$$

Let us express m ч:

$$m = \frac{k_1}{\frac{a_0 f}{a_0 - f} - (a_0 + f)}$$

In total:

$$m = 21 \text{ mm}$$

$$\Delta m = 3 \text{ mm}$$

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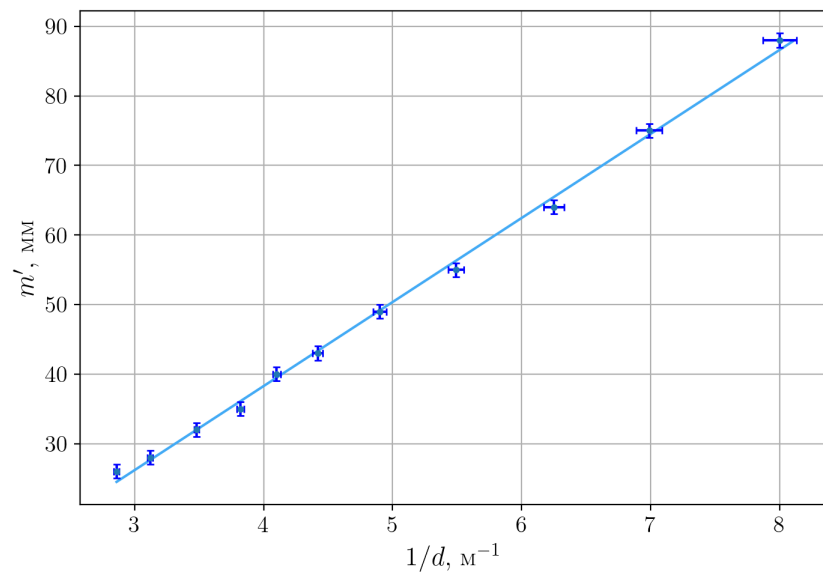
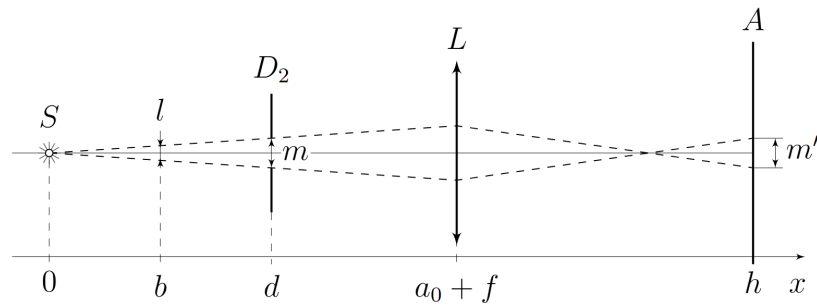
Total:

$$m = 21 \text{ mm}$$

$$\Delta m = 3 \text{ mm}$$

Answer: \textbf{Answer:}

$$m = (21 \pm 3) \text{ mm}$$



C1

1.80 pts

As you move the lens, the appearance of the image on the screen will change. Vary the position of the lens (i.e. the value of a) and for each a sketch the appearance of the image on your answer sheets. Your image should clearly indicate (by shading or otherwise) highlight, shadow, and penumbra (if any).

Commentary on pictures. $a = 210$ mm - focused LED; $a = 250$ mm - blurry circle (almost a square); $a = 300$ mm - focused square (stripes along the edges are reflections from the edges); $a = 365$ mm - cross, almost focused, cut into a square; $a = 470$ mm - cross, cut into a square, blurred; $a = 720$ mm - a cross cut off by a circle (lens), very blurred.

Commentary on pictures.